

GOVERNMENT OF INDIA • MINISTRY OF CONSUMER AFFAIRS, FOOD & PUBLIC DISTRIBUTION

DEPARTMENT OF CONSUMER AFFAIRS — DIRECTORATE OF LEGAL METROLOGY

NATIONAL LEGAL METROLOGY TYPE EVALUATION LABORATORY (OIML ISSUING AUTHORITY)

PATTERN EVALUATION TEST REPORT • NON-AUTOMATIC WEIGHING INSTRUMENT (NAWI)

Evaluated in accordance with OIML Recommendation R 76-1:2006 (E) & Legal Metrology Act, 2009

Report No:	OIML-R76-2026-IND-0004	Application No:	APP-NAWI-2026-0004
Testing Date:	2026-09-29	Overall Finding:	DRAFT

1. INSTRUMENT IDENTIFICATION & TECHNICAL SPECIFICATIONS

Manufacturer:	Bharat Precision Metrology	Instrument Type:	Non-Automatic Bench Scale
Manufacturer Address:	Okhla Industrial Area, Phase III, New Delhi	Model Name:	BPM-30-ADV
Accuracy Class:	Class III	Serial Number:	SN-2026-00004
Max Capacity (Max):	30.0 kg	Scale Interval (d):	0.01 kg
Min Capacity (Min):	0.2 kg	Verification Interval (e):	0.01 kg
Verification Scale Intervals (n):	n = 3000	Tare Subtraction (Max T):	-30.0 kg
Operating Temperature:	-10.0°C to +40.0°C	Power Supply:	230V AC (+10% / -15%), 50Hz
Load Cell & Indicator:	None	Firmware / Checksum:	v1.0.0 (None)

2. LABORATORY AND ENVIRONMENTAL CONDITIONS (Clause A.3)

Testing Laboratory:	National Legal Metrology Type Evaluation Laboratory	Ambient Temperature:	23.5 °C
Laboratory Accreditation:	NABL ISO/IEC 17025 Accredited & OIML Issuing Authority	Relative Humidity:	52.0 % RH
Local Gravity Acceleration (g):	9.7803 m/s²	Barometric Pressure:	1013.2 hPa
Standard Calibration Weights:	Class M1 & F2 Working Standards (Traceable to NPL India)	Testing Officer:	Er. Rajesh Sharma (Senior Metrological Officer)

3. WEIGHING PERFORMANCE TEST OBSERVATIONS & EVALUATION (Clause A.4.4)

#	Dir	Load (kg)	Indication I	ΔL (kg)	Error E _c	E _c (e)	MPE (±e)	MPE (kg)	Verdict
1	INCR	0	0	0.005	+0.0000	+0.00 e	±0.5 e	±0.0025	PASS
2	INCR	10	10	0.005	+0.0000	+0.00 e	±1.0 e	±0.0050	PASS
3	INCR	30	30	0.005	+0.0000	+0.00 e	±1.5 e	±0.0075	PASS
4	DEC R	10	10	0.005	+0.0000	+0.00 e	±1.0 e	±0.0050	PASS
5	DEC R	0	0	0.005	+0.0000	+0.00 e	±0.5 e	±0.0025	PASS

4. COMPREHENSIVE OIML R 76-1 TEST COMPLIANCE SUMMARY

Clause	Test Procedure	Statutory Acceptance Criteria	Result Status
Clause A.4.4	Weighing Performance Test (Incr/Decr)	Corrected error E _c ≤ MPE across range; Hysteresis ≤ MPE	PASS
Clause A.4.10	Repeatability Test (at 0.5 Max & Max)	Max difference between weighings ≤ MPE for load	PASS

Clause A.4.7	Eccentricity / Off-Center Loading	$ E_c \leq \text{MPE at } 1/3 \text{ Max at 4 corners and center}$	PASS
Clause A.4.2	Zero-Setting and Zero-Tracking	Residual zero error $\leq \pm 0.25 e$	PASS
Clause A.4.8	Discrimination Test	Extra load 1.4 d gives noticeable change of indication	PASS
Clause A.5.3	Static Temperatures (-10°C to +40°C)	Span error $\leq \text{MPE}$; Zero drift $\leq 1 e / 5^\circ\text{C}$	PASS
Clause A.5.4	Voltage Variations (+10% / -15%)	Indications remain within MPE limits	PASS

STATUTORY MODEL APPROVAL VERDICT: COMPLIANT & APPROVED

The weighing instrument submitted meets all metrological and technical requirements of OIML Recommendation R 76-1:2006 (E) and the Legal Metrology (General) Rules, 2011. Type approval is hereby recommended under Section 22 of the Legal Metrology Act, 2009.

TESTED & EVALUATED BY: Er. Rajesh Sharma (Senior Metrological Officer) Senior Metrological Officer, Legal Metrology Digital Signature: VERIFIED • ID: GOV-IN-LM-0881 Timestamp: 2026-09-29 10:28:34 UTC SHA-256 Hash: 9f86d081884c7d659a2feaa0c55ad015...	VERIFIED & APPROVED BY: Dr. Priya V. Iyer (Director of Legal Metrology) Director of Legal Metrology, Government of India Official Stamp: [DIGITALLY SIGNED & VERIFIED] Legal Metrology Act, 2009 Statutory Repository Tamper-Evident Digital Certification Authority
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